AI Programming Foundations

C2 • Titanic data workflow

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Overview

This project builds an executable Python workflow for loading, cleaning, exploring, and visualizing Titanic passenger records. The supplied dataset is the Data Science Dojo Titanic CSV (Data Science Dojo, n.d.; https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv). The deliverables include a saved notebook, dependency file, run instructions, and this report; findings are descriptive rather than causal.

Dataset Description

The raw CSV contains 891 passenger records and 12 columns, with 0 duplicate rows. Seven columns are numeric (Age and Fare are floats; the other five are integers), and five contain text. The outcome is Survived, encoded as 0 or 1; key explanatory fields are Pclass, Sex, Age, and the family counts SibSp and Parch. Cabin has 687 missing values, Age has 177, and Embarked has 2; no other column has missing values. PassengerId, Name, and Ticket are retained without transformation, and no claim is made that these records represent every person aboard or a modern population.

Workflow Description

The notebook imports NumPy, pandas, Matplotlib, and Seaborn and fixes the NumPy seed at 42. It loads the local CSV, displays its first rows and shape, and audits duplicates, types, and missingness. Two documented cleaning functions preserve the raw frame, fill missing values, and derive family features; a documented EDA function returns group counts and survival rates. Three numbered plots and a closing interpretation complete the workflow, with assertions checking row preservation, valid outcomes, and missing-value handling.

Key Decisions and Assumptions

The overall observed Age median of 28 years fills 177 gaps while retaining every passenger; the median is less sensitive to extreme ages than the mean, but it compresses variation. AgeWasMissing preserves which ages were filled. Unknown labels for Cabin and Embarked preserve missingness without inventing cabins or ports, and FamilySize = SibSp + Parch + 1 counts the passenger plus recorded relatives; IsAlone marks FamilySize = 1. EDA focuses on class, recorded sex, and family status with denominators shown. Figures 1 and 2 compare survival rates using all rows, while Figure 3 compares normalized distributions of observed ages only, avoiding an artificial median spike.

Results and Interpretation

Overall, 342 of 891 passengers survived (38.4%). Figure 1 shows survival of 63.0% in first class (136/216), 47.3% in second class (87/184), and 24.2% in third class (119/491). Figure 2 shows 74.2% survival among recorded female passengers (233/314), compared with 18.9% among recorded male passengers (109/577). Passengers alone had 30.4% survival (163/537), compared with 50.6% for those with recorded relatives aboard (179/354). Figure 3 shows overlapping age distributions concentrated in young adulthood; its within-outcome density scaling compares shapes rather than total survivor counts. These associations can overlap and are not estimates of the causal effects of class, sex, family status, or age.

Responsible Practice

Age is missing for 19.9% of records, and deleting those 177 rows can change the composition of the analyzed population. The notebook compares complete observed ages, overall median filling, and medians within class and recorded sex using an under-18 threshold, retaining counts and rates for each method. These treatments can shift age-related conclusions; filling Age alone cannot alter class or sex survival rates when all rows are retained. Single median filling neither models uncertainty nor establishes that missingness is random; assumptions and sensitivity checks matter in missing-data analysis (Sterne et al., 2009). Cabin is missing for 77.1% of passengers, so an Unknown label identifies absent information rather than recovering actual cabin identities. Figure 3 excludes missing ages and is therefore potentially selective despite avoiding fabricated observations. Before modeling, compare missingness and error rates across groups, keep sensitive fields under deliberate review, and avoid treating historical disparities or the binary recorded sex field as universal or prescriptive.

Reproducibility

The local CSV, seven-section notebook with saved outputs, figures, requirements.txt, and README.md are versioned together. Run the notebook from its repository directory in a Python 3.12 virtual environment after installing the frozen runtime dependencies and the separate notebook tools described in README.md; its transformations are deterministic and the random seed is fixed at 42. Versioned data, executable explanations, and recorded dependencies support transparent reruns, consistent with Danchev (2022). Work was committed incrementally on development and integrated into main, with AI assistance identified in the Git author metadata. The project repository is https://github.com/Mohammed-MM3/ai-programming-foundations-project. Both main and development preserve the incremental project history; publication is checked separately in VERIFICATION.md.

Future Integration Reflections

For machine learning, I would split the 891 rows into stratified training, validation, and test sets before fitting the Age median or encoders, preventing leakage from held-out data. I would encode Unknown categories, retain AgeWasMissing, check redundancy between FamilySize and IsAlone, and compare a simple baseline with any more complex model. Before a neural network, I would scale numeric features using training statistics, check class and recorded-sex imbalance, and question whether 891 rows justify its capacity. Agentic automation could rerun validation, regenerate figures, and detect schema changes, but imputation choices, fairness conclusions, and any publication should remain subject to human review. Such automation must not silently infer cabins from 687 missing values or treat filled ages as observations.

Age sensitivity detail

Under-18 survival is 54.0% (61/113) under all three age treatments. Adult survival is 38.1% (229/601) among observed ages and 36.1% (281/778) after either filling method. The two filling methods yield the same age-group allocation here, correcting the brief's expectation that every comparison must differ. Mean ages among non-survivors are 30.63 years for observed ages, 30.03 after overall-median filling, and 29.74 after class/sex filling; survivor means are 28.34, 28.29, and 28.11 years, respectively. Thus some age statistics change while the chosen child/adult grouping does not distinguish the two fill methods.

Figures referenced in the notebook

Figure 1. Survival rate by passenger class

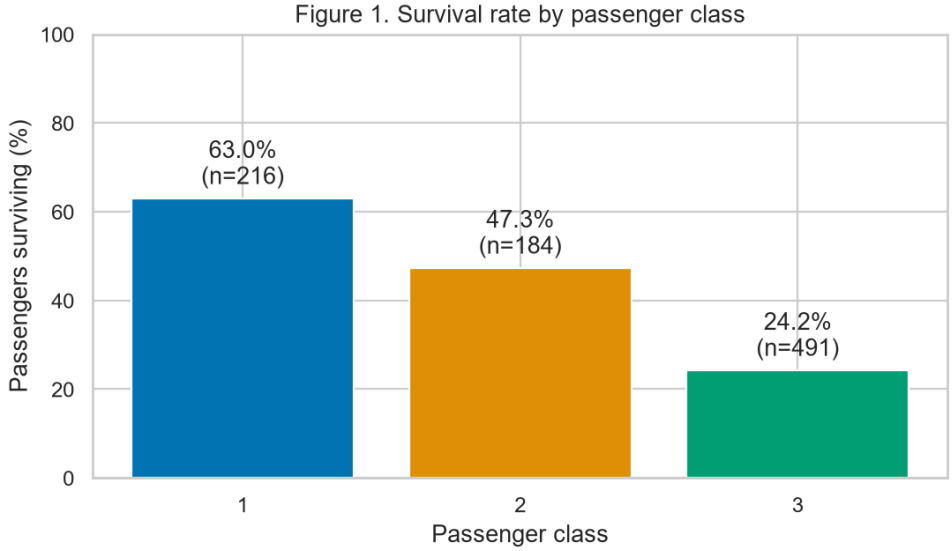


Figure 2. Survival rate by recorded sex

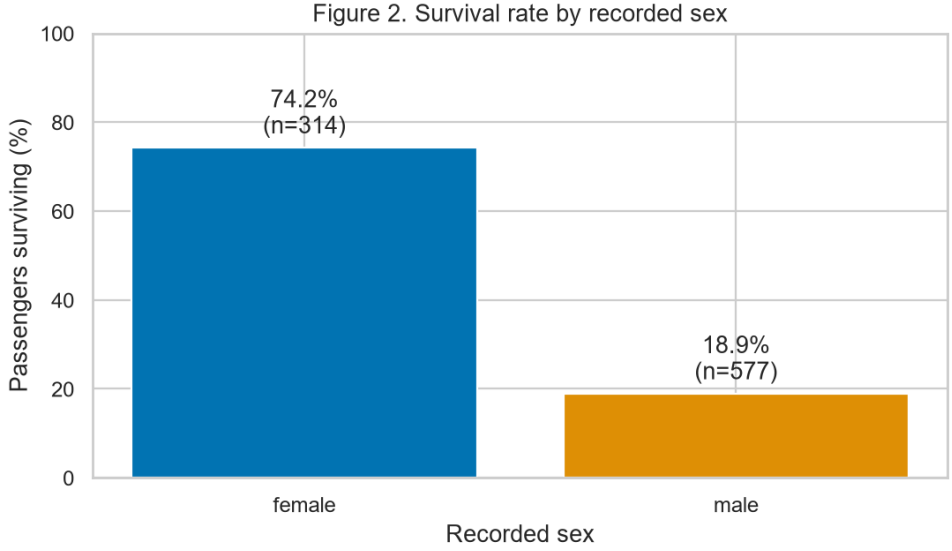
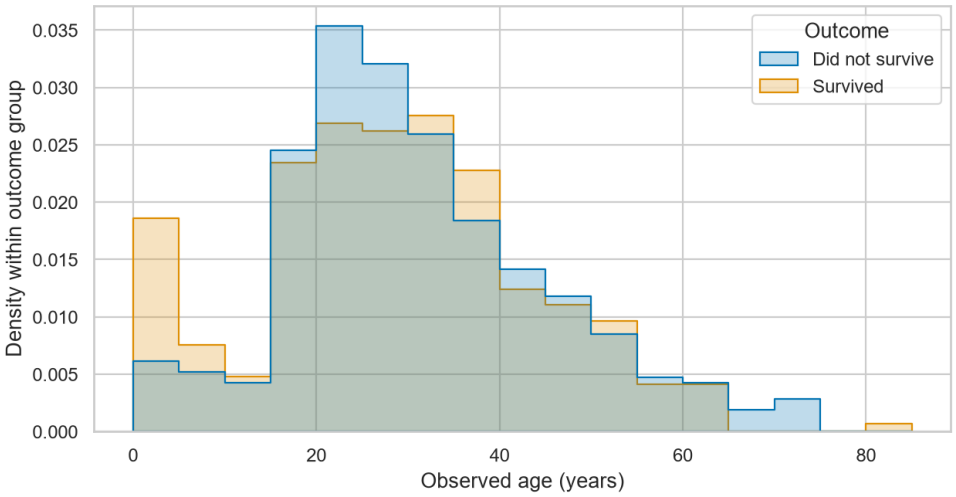


Figure 3. Observed age distribution by survival outcome



Sources and Citations

References

Danchev, V. (2022). Reproducible data science with Python: An open learning resource. *Journal of Open Source Education, 5*(56), Article 156. https://doi.org/10.21105/jose.00156

Data Science Dojo. (n.d.). *Titanic passenger dataset* [Data set]. GitHub. https://github.com/datasciencedojo/datasets/blob/master/titanic.csv

Sterne, J. A. C., White, I. R., Carlin, J. B., Spratt, M., Royston, P., Kenward, M. G., Wood, A. M., & Carpenter, J. R. (2009). Multiple imputation for missing data in epidemiological and clinical research: Potential and pitfalls. *BMJ, 338*, Article b2393. https://doi.org/10.1136/bmj.b2393

Final interpretation

The strongest descriptive differences are the class and recorded-sex survival gaps. Family status is also associated with survival, while missing ages make age-based comparisons dependent on data handling. Retaining rows, documenting missingness, and exposing the sensitivity analysis make the workflow useful as a foundation for later modeling, subject to leakage controls, group evaluation, and human review.